



Qualification Pack



Scaffold Design Engineer

QP Code: SSD/VSQ/Q0203

Version: 1.0

NSQF Level: 6

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SSD/VSQ/Q0203: Scaffold Design Engineer

Brief Job Description

A Scaffold Design Engineer is responsible for creating safe and efficient scaffold designs using industry standards and codes utilizes scaffold structural design software and computer-aided design (CAD) & systems to develop accurate scaffold drawings. The design engineer calculates loads in accordance with national & international standards and analyzes scaffold structures using design software like STAAD Pro software and have good understanding of scaffold specifications, drawings, Indian and international standard codes to ensure compliance and safety in scaffold designs.

Personal Attributes

He should be physically & mentally fit and should be able to provide design advice on the suitability of specialized scaffolds to meet the health and safety requirements regarding design and technical advice on scaffolding works.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSD/VSQ/N0213: Scaffoldings & Specifications.](#)
2. [SSD/VSQ/N0214: Understanding Scaffold Drawings & Designs, Indian & International Standard Codes](#)
3. [SSD/VSQ/N0215: Scaffold Design & Drawings using the scaffold & Computer-Aided Design \(CAD\) system](#)
4. [SSD/VSQ/N0216: Calculation of loads in scaffold designs as per Indian & International Standard](#)
5. [SSD/VSQ/N0217: Analysis of Scaffold design using STAAD Pro as per applicable IS and International Codes](#)
6. [SSD/VSQ/N0218: Plan, Organise & Monitor Scaffolding Safety Protocols](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Cross Sectoral
Sub-Sector	Construction, Infrastructure, Real estate, Iron & Steel, Mining, Logistics, Hydrocarbon and Others.



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Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
Country	India
NSQF Level	6
Credits	23
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2141.0100 ; Works Inspector, Engineering /Inspection Engineer
Minimum Educational Qualification & Experience	Completed 4 year UG program (in relevant field) with 2 Years of experience OR Completed 3 year UG degree (in relevant field) with 3 Years of experience OR Previous relevant Qualification of NSQF Level (5.5) with 1-2 Years of experience (1.5 Years) OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	Nil
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	27/08/2027
NSQC Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QC-06-CO-02971-2024-V1-SSDF
NQR Version	1.0

Remarks:

NA



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SSD/VSQ/N0213: Scaffoldings & Specifications.

Description

The NOS describes the knowledge & skills required by the professional to know & understand the types of scaffoldings, their components, specifications, uses under specific conditions and protections required for safe use.

Scope

The scope covers the following :

- Identification of scaffold & components.
- Design load calculation of the scaffold.
- Fall protection requirements & provisions in the scaffold.

Elements and Performance Criteria

Types of Scaffoldings

To be competent, the user/individual on the job must be able to:

- PC1.** Identify various types of scaffolds, their components.
- PC2.** Determine type of scaffold required as per site & load requirements.
- PC3.** Identify working & faulty components and defect in the components.

Design Load Calculation

To be competent, the user/individual on the job must be able to:

- PC4.** Calculate load on scaffold & optimum load.
- PC5.** Calculate design load for the scaffold.
- PC6.** Analyse Working requirements of components, tie-offs, supports etc. of the scaffoldings as per design requirement.

Fall Protection Requirements

To be competent, the user/individual on the job must be able to:

- PC7.** Identify types of fall protection for the scaffolds, tie-offs, supports and ladders.
- PC8.** Work out fall protections required in the scaffold for various activities and effectiveness.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Several types of scaffoldings used under various working conditions.
- KU2.** Components of scaffolds and their uses.
- KU3.** Find faulty scaffold components.
- KU4.** Determination of load classes of scaffolds, design loads and optimum load for the scaffold.
- KU5.** Several kind of fall protection provided, tie-offs, supports, ladders of the scaffolds.
- KU6.** Purpose of each of the components and fall protection in the scaffolds.



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KU7. Good practices while erecting the scaffold and while using the scaffold.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Basic mathematics and calculations.
- GS3.** Understanding & adherence of instructions.
- GS4.** Understanding of safety and adherence to safety related protocols.
- GS5.** Conduct of Professional behavior with co-workers and employees.
- GS6.** Planning of work & resource optimisation.
- GS7.** Understanding & use of personal protective equipments.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Organizational hierarchy for availability of tools, equipment & material & its security.
- GS11.** Ready to learn, find solution, decision making & conflict resolution.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Types of Scaffoldings</i>	21	14	-	-
PC1. Identify various types of scaffolds, their components.	9	6	-	-
PC2. Determine type of scaffold required as per site & load requirements.	6	4	-	-
PC3. Identify working & faulty components and defect in the components.	6	4	-	-
<i>Design Load Calculation</i>	21	14	-	-
PC4. Calculate load on scaffold & optimum load.	6	4	-	-
PC5. Calculate design load for the scaffold.	6	4	-	-
PC6. Analyse Working requirements of components, tie-offs, supports etc. of the scaffoldings as per design requirement.	9	6	-	-
<i>Fall Protection Requirements</i>	18	12	-	-
PC7. Identify types of fall protection for the scaffolds, tie-offs, supports and ladders.	9	6	-	-
PC8. Work out fall protections required in the scaffold for various activities and effectiveness.	9	6	-	-
NOS Total	60	40	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0213
NOS Name	Scaffoldings & Specifications.
Sector	Cross Sectoral
Sub-Sector	
Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



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SSD/VSQ/N0214: Understanding Scaffold Drawings & Designs, Indian & International Standard Codes

Description

The NOS describes the knowledge & skills required by the professional to know & understand the scaffolding drawings & design, factors influencing design of scaffold, load calculations, design & safety parameters for supported, un-supported, cantilever, mobile, suspended scaffoldings.

Scope

The scope covers the following :

- Reading & Understanding scaffold drawings.
- Scaffold parameters for design & safety.
- Identification & selection of fall protection & design selection.

Elements and Performance Criteria

Understanding Scaffold Drawing & Details

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret scaffold drawings & safety elements.
- PC2.** Interpret scaffold drawings & designs.
- PC3.** Interpret details of scaffold drawings.

Scaffold Parameters for Design & Safety

To be competent, the user/individual on the job must be able to:

- PC4.** Work out design & safety parameters of scaffolds as per Indian Standards IS-2750 & IS-3696.
- PC5.** Work out design details of scaffold as per International Standards of OSHA & BS standards & safety parameters.
- PC6.** Check design details of scaffolds and its interpretations.

Identification & selection of fall protection & design

To be competent, the user/individual on the job must be able to:

- PC7.** Identify fall protections & design for fall protection.
- PC8.** Work out details of ladder/temporary ladder requirements & design.
- PC9.** Analyse factors affecting designing of scaffold.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Drawings of scaffolds and its interpretation.
- KU2.** Details required of scaffold drawing preparation.
- KU3.** Load calculations, classification and design of various types of scaffoldings & its components.
- KU4.** Indian & International design parameters & standards of scaffolds and design loads.



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- KU5.** Making design checks of scaffolds design details provided to him.
- KU6.** Fall protection requirements and fall protection measures.
- KU7.** Requirements of ladders/temporary ladders.
- KU8.** Design & drawing software operation.
- KU9.** Knowledge of safety rules, safety measures, precautions and safety violating actions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Knowledge of drawing & design software.
- GS3.** Understanding & adherence of instructions.
- GS4.** Understanding of safety and adherence to safety related protocols.
- GS5.** Conduct of Professional behavior with co-workers and employees.
- GS6.** Planning of work & resource optimisation.
- GS7.** Understanding & use of personal protective equipments.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Organizational hierarchy for availability of tools, equipment & material & its security.
- GS10.** Ready to learn, find solution, decision making & conflict resolution.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Scaffold Drawing & Details</i>	21	14	-	-
PC1. Interpret scaffold drawings & safety elements.	9	6	-	-
PC2. Interpret scaffold drawings & designs.	6	4	-	-
PC3. Interpret details of scaffold drawings.	6	4	-	-
<i>Scaffold Parameters for Design & Safety</i>	21	14	-	-
PC4. Work out design & safety parameters of scaffolds as per Indian Standards IS-2750 & IS-3696.	9	6	-	-
PC5. Work out design details of scaffold as per International Standards of OSHA & BS standards & safety parameters.	6	4	-	-
PC6. Check design details of scaffolds and its interpretations.	6	4	-	-
<i>Identification & selection of fall protection & design</i>	18	12	-	-
PC7. Identify fall protections & design for fall protection.	6	4	-	-
PC8. Work out details of ladder/temporary ladder requirements & design.	6	4	-	-
PC9. Analyse factors affecting designing of scaffold.	6	4	-	-
NOS Total	60	40	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0214
NOS Name	Understanding Scaffold Drawings & Designs, Indian & International Standard Codes
Sector	Cross Sectoral
Sub-Sector	
Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



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SSD/VSQ/N0215: Scaffold Design & Drawings using the scaffold & Computer-Aided Design (CAD) system

Description

This NOS covers and describes the skills and knowledge required to prepare dimensional drawings of scaffolding using structural design software & computer-aided design (CAD) system. The scope covers the creation and correction of 2/3 Dimensional Drawings that are used in all projects of construction across all sub-sectors using CAD systems.

Scope

The scope covers the following :

- Calculation of dimensions of components & complete the drawing.
- Preparation of scaffold drawings & highlighting the components, specification & critical feature.
- Creating table and symbols for the drawing.

Elements and Performance Criteria

Calculation of dimensions of components & completing drawings

To be competent, the user/individual on the job must be able to:

- PC1.** Carry out necessary calculations to compute dimensions of various components/ parts of drawings.
- PC2.** Convert the dimensions to the required scale to input in the system.
- PC3.** Use drafting principles to produce cad drawings showing plans, sections, elevations, and different types of views.
- PC4.** Use appropriate commands in the software to draw the required drawings as per standard practices.

Preparation of scaffold drawings & highlighting the components, specification & critical feature

To be competent, the user/individual on the job must be able to:

- PC5.** Use keyboard commands and pull-down menus available in common CAD systems to prepare the drawings.
- PC6.** Use codes and other references that follow the required conventions.
- PC7.** Draw structures to highlight critical features in accordance with specifications and requirements.
- PC8.** Draw scaffold assemblies to highlight critical features as per specification.

Creating table and symbols for the drawing

To be competent, the user/individual on the job must be able to:

- PC9.** Create tables to denote the name, dimensions, perimeter, and area of various parts or components as per client requirement.
- PC10.** Use relevant and appropriate symbols as per drawing requirements to provide details in the drawings.

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** Drawings of scaffolds and their interpretation for scaffolds.
- KU2.** Details required of scaffold drawing preparation for scaffolds.
- KU3.** Load calculations, classification, and design of various types of scaffoldings & their components.
- KU4.** Indian & International design parameters & standards of scaffolds and design loads for cantilever & suspended scaffolds.
- KU5.** Making design checks of scaffolds design details for cantilever & suspended scaffolds.
- KU6.** Fall protection requirements and fall protection measures for scaffolds including cantilever & suspended scaffolds.
- KU7.** Requirements of ladders/temporary ladders for cantilever & suspended scaffolds.
- KU8.** Design & drawing software operation.
- KU9.** Knowledge of safety rules, safety measures, precautions and safety-violating actions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Knowledge of drawing & design software.
- GS3.** Understanding & adherence of instructions.
- GS4.** Understanding of safety and adherence to safety related protocols.
- GS5.** Conduct of Professional behavior with co-workers and employees.
- GS6.** Planning of work & resource optimisation.
- GS7.** Understanding & use of personal protective equipments.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Organizational hierarchy for availability of tools, equipment & material & its security.
- GS10.** Ready to learn, find solution, decision making & conflict resolution.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Calculation of dimensions of components & completing drawings</i>	24	16	8	-
PC1. Carry out necessary calculations to compute dimensions of various components/ parts of drawings.	6	4	-	-
PC2. Convert the dimensions to the required scale to input in the system.	6	4	-	-
PC3. Use drafting principles to produce cad drawings showing plans, sections, elevations, and different types of views.	6	4	-	-
PC4. Use appropriate commands in the software to draw the required drawings as per standard practices.	6	4	-	-
<i>Preparation of scaffold drawings & highlighting the components, specification & critical feature</i>	24	16	8	-
PC5. Use keyboard commands and pull-down menus available in common CAD systems to prepare the drawings.	6	4	-	-
PC6. Use codes and other references that follow the required conventions.	6	4	-	-
PC7. Draw structures to highlight critical features in accordance with specifications and requirements.	6	4	-	-
PC8. Draw scaffold assemblies to highlight critical features as per specification.	6	4	-	-
<i>Creating table and symbols for the drawing</i>	12	8	4	-
PC9. Create tables to denote the name, dimensions, perimeter, and area of various parts or components as per client requirement.	6	4	-	-
PC10. Use relevant and appropriate symbols as per drawing requirements to provide details in the drawings.	6	4	-	-
NOS Total	60	40	20	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0215
NOS Name	Scaffold Design & Drawings using the scaffold & Computer-Aided Design (CAD) system
Sector	Cross Sectoral
Sub-Sector	
Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



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SSD/VSQ/N0216: Calculation of loads in scaffold designs as per Indian & International Standard

Description

The NOS describes the knowledge & skills required by the professionals about scaffolding design load calculation as per IS-875 & IS-3696, international practices and various codal provisions followed in designing of scaffolds as per specifications.

Scope

The scope covers the following :

- Dead Loads, imposed loads, wind loads as per Indian Standard code of practice for design loads.
- Load combinations for design loads.
- International practices in the design of scaffolds.

Elements and Performance Criteria

Understand Load on Scaffold

To be competent, the user/individual on the job must be able to:

- PC1.** Identify loads & load factors affecting scaffold design.
- PC2.** Comprehend design load calculation on scaffold.
- PC3.** Analyse dead loads, imposed loads and wind loads on scaffold.

Understand Codes for Load Calculation

To be competent, the user/individual on the job must be able to:

- PC4.** Analyse load IS-875 and its element applicable for scaffolds.
- PC5.** Analyse provisions of IS-3696.
- PC6.** Apply IS-875 & IS-3696 in load calculation and international codes & practices.

Load calculation on Scaffold

To be competent, the user/individual on the job must be able to:

- PC7.** Calculate dead load details of scaffold as per Indian Standard code for scaffold.
- PC8.** Calculate imposed loads & wind loads as per Indian & International standard codes for scaffolds.
- PC9.** Calculate combination of loads on scaffold.
- PC10.** Check design load details of the scaffold as per international design standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Types of scaffoldings & its components.
- KU2.** Load calculation, safety provisions as per international practice.
- KU3.** Design practices as per international codes.



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- KU4.** Design checks of scaffolds as per international practices & convention.
- KU5.** Scaffold drawings as per international concept & practices.
- KU6.** Protections are to be provided in the scaffold.
- KU7.** Requirements of ladders/temporary ladders.
- KU8.** Design & drawing software operation.
- KU9.** Safety rules, safety measures, precautions and safety violating actions.
- KU10.** Inspection process & documents preparation.
- KU11.** Reporting process of inspection report.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Knowledge of drawing & design software.
- GS3.** Understanding & adherence of instructions.
- GS4.** Understanding of safety and adherence to safety related protocols.
- GS5.** Conduct of Professional behavior with co-workers and employees.
- GS6.** Planning of work & resource optimisation.
- GS7.** Understanding & use of personal protective equipments.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Organizational hierarchy for availability of tools, equipment & material & its security.
- GS10.** Ready to learn, find solution, decision making & conflict resolution.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand Load on Scaffold</i>	18	12	3	-
PC1. Identify loads & load factors affecting scaffold design.	6	4	-	-
PC2. Comprehend design load calculation on scaffold.	6	4	-	-
PC3. Analyse dead loads, imposed loads and wind loads on scaffold.	6	4	-	-
<i>Understand Codes for Load Calculation</i>	18	12	3	-
PC4. Analyse load IS-875 and its element applicable for scaffolds.	6	4	-	-
PC5. Analyse provisions of IS-3696.	6	4	-	-
PC6. Apply IS-875 & IS-3696 in load calculation and international codes & practices.	6	4	-	-
<i>Load calculation on Scaffold</i>	24	16	4	-
PC7. Calculate dead load details of scaffold as per Indian Standard code for scaffold.	6	4	-	-
PC8. Calculate imposed loads & wind loads as per Indian & International standard codes for scaffolds.	6	4	-	-
PC9. Calculate combination of loads on scaffold.	6	4	-	-
PC10. Check design load details of the scaffold as per international design standards.	6	4	-	-
NOS Total	60	40	10	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0216
NOS Name	Calculation of loads in scaffold designs as per Indian & International Standard
Sector	Cross Sectoral
Sub-Sector	
Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

SSD/VSQ/N0217: Analysis of Scaffold design using STAAD Pro as per applicable IS and International Codes

Description

The NOS covers the knowledge & skills required by the professional in the analysis of scaffolding using the software STAAD Pro as per applicable IS & International codes. Scaffold drawings, design & various codes provisions, specifications & best practices of scaffoldings.

Scope

The scope covers the following :

- Design check and analysis of scaffolding using STAAD Pro.
- Safety measures in the scaffold are to be covered during the design of the scaffold.
- Documents to be prepared and maintained in scaffold design.

Elements and Performance Criteria

Analysis of Scaffolding using STAAD Pro as per applicable IS and International Codes

To be competent, the user/individual on the job must be able to:

- PC1.** Draw structures to coordinates, nodes, dimensions and specification.
- PC2.** Draw scaffold assemblies to highlight critical features as per specification.
- PC3.** Apply loads on scaffold as per calculations.
- PC4.** Carry out the scaffold design in software for complete analysis.
- PC5.** Determine the required scaffold components and their placement based on design calculations and load requirements.
- PC6.** Ensure compliance with relevant codes and standards throughout the design and analysis process.

Reading Scaffold Analysis data and details

To be competent, the user/individual on the job must be able to:

- PC7.** Read and understand the analysed results and understand the outcome.
- PC8.** Extract the required result and prepare the check list for each component.
- PC9.** Check critical nodes and each of the inspection points of the scaffold.

Drawings & Documentation

To be competent, the user/individual on the job must be able to:

- PC10.** Prepare & extract the software drawing after analysis.
- PC11.** Prepare and extract result data after analysis for checking & records.
- PC12.** Prepare the data sheet and drawing sheet extracted from the software after analysis for records.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** Design of various types of scaffoldings & their components.
- KU2.** International design standards of scaffolds & design loads.
- KU3.** Computer uses in reading drawings & designs.
- KU4.** Communications on drawings & designs.
- KU5.** Design checks of scaffolds as per international practices & conventions.
- KU6.** Drawings of scaffolds and its interpretation.
- KU7.** Details required of scaffold drawing preparation.
- KU8.** Load calculations, classification and design of various types of scaffoldings & its components.
- KU9.** Indian & International design parameters & standards of scaffolds and design loads.
- KU10.** Making design checks of scaffolds design details provided to him.
- KU11.** Fall protection requirements and fall protection measures.
- KU12.** Requirements of ladders/temporary ladders.
- KU13.** Design & drawing software operation.
- KU14.** Safety rules, safety measures, precautions, and safety-violating actions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculating tools.
- GS4.** Computer operation, email writing, documents preparation.
- GS5.** Knowledge of operation MS Excel sheet.
- GS6.** Computing/mathematical software operation on computer.
- GS7.** Understanding complexities, priorities & severity and reporting method as per organizational hierarchy.
- GS8.** Planning, organising & resource optimisation.
- GS9.** Administration, decision making & conflict resolution.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analysis of Scaffolding using STAAD Pro as per applicable IS and International Codes</i>	36	24	12	-
PC1. Draw structures to coordinates, nodes, dimensions and specification.	6	4	-	-
PC2. Draw scaffold assemblies to highlight critical features as per specification.	6	4	-	-
PC3. Apply loads on scaffold as per calculations.	6	4	-	-
PC4. Carry out the scaffold design in software for complete analysis.	6	4	-	-
PC5. Determine the required scaffold components and their placement based on design calculations and load requirements.	6	4	-	-
PC6. Ensure compliance with relevant codes and standards throughout the design and analysis process.	6	4	-	-
<i>Reading Scaffold Analysis data and details</i>	12	8	4	-
PC7. Read and understand the analysed results and understand the outcome.	5	3	-	-
PC8. Extract the required result and prepare the check list for each component.	4	2	-	-
PC9. Check critical nodes and each of the inspection points of the scaffold.	3	3	-	-
<i>Drawings & Documentation</i>	12	8	4	-
PC10. Prepare & extract the software drawing after analysis.	5	3	-	-
PC11. Prepare and extract result data after analysis for checking & records.	4	2	-	-
PC12. Prepare the data sheet and drawing sheet extracted from the software after analysis for records.	3	3	-	-
NOS Total	60	40	20	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0217
NOS Name	Analysis of Scaffold design using STAAD Pro as per applicable IS and International Codes
Sector	Cross Sectoral
Sub-Sector	
Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
NSQF Level	6
Credits	5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



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SSD/VSQ/N0218: Plan, Organise & Monitor Scaffolding Safety Protocols

Description

The NOS describes the knowledge & skills required by the professionals about planning, organizing and monitoring of their work to provide the expected outcomes efficiently & ensuring quality of the work.

Scope

The scope covers the following :

- Planning of resources for own work and communication to concerned subordinates, co-workers and superiors.
- Provide necessary support to subordinates, co-ordinate with co-workers and liaise with superiors and other teams.
- Monitor progress of work and adjust, manage or project requirements on time.

Elements and Performance Criteria

Planning of Work

To be competent, the user/individual on the job must be able to:

- PC1.** Plan the resources, schedules and timelines as per work timelines given by superiors.
- PC2.** Understand hierarchy of the organisation and communicate to concerned co-workers & superiors.
- PC3.** Task the subordinates as per task & timelines.

Organising of Work

To be competent, the user/individual on the job must be able to:

- PC4.** Organise resource collection and provisioning.
- PC5.** Communicate & Organise effective communication to concerned co-workers & superiors.
- PC6.** Brief the subordinates about the schedule, sequence, timing & resources.

Monitoring of Work

To be competent, the user/individual on the job must be able to:

- PC7.** Monitor progress of work, management of resources, guidance to subordinates and monitoring tools.
- PC8.** Learn reporting procedure & tools, report preparation to superiors and keeping the other teams informed.
- PC9.** Analyse Documentations methods, record maintenance, report submission & compliances.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Task, roles & responsibility of own and subordinates.
- KU2.** Planning timelines of task, sequence and resources deployment for the task.



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- KU3.** Roles & tasks of co-workers & other teams working for the same task.
- KU4.** Instruction, advices, briefing and resource allocation to subordinates.
- KU5.** Communication to co-workers, other teams & superiors for the work.
- KU6.** Organising tools, equipment, materials & other resources on time as per timelines & task.
- KU7.** Providing technical support to his team & subordinates.
- KU8.** Planning & monitoring tools & software and its operation.
- KU9.** Complexities, priorities & severity of the task.
- KU10.** Report preparation & reporting methods as per organizational hierarchy & reports submission.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculating tools.
- GS4.** Planning, organising & resource optimisation.
- GS5.** Administration, decision making & conflict resolution.
- GS6.** Planning & monitoring tools, software operation.
- GS7.** Understanding & adherence of instructions.
- GS8.** Understanding of safety and adherence to safety related protocols.
- GS9.** Conduct of Professional behavior with co-workers and employees.
- GS10.** Use of personal protective equipments.
- GS11.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS12.** Housekeeping and disposal of waste materials.
- GS13.** Organizational hierarchy for availability of tools, equipment & material & its security.
- GS14.** Ready to learn, understand, find solution & conflict resolution.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Planning of Work</i>	24	16	-	-
PC1. Plan the resources, schedules and timelines as per work timelines given by superiors.	12	8	-	-
PC2. Understand hierarchy of the organisation and communicate to concerned co-workers & superiors.	6	4	-	-
PC3. Task the subordinates as per task & timelines.	6	4	-	-
<i>Organising of Work</i>	18	12	-	-
PC4. Organise resource collection and provisioning.	6	4	-	-
PC5. Communicate & Organise effective communication to concerned co-workers & superiors.	6	4	-	-
PC6. Brief the subordinates about the schedule, sequence, timing & resources.	6	4	-	-
<i>Monitoring of Work</i>	18	12	-	-
PC7. Monitor progress of work, management of resources, guidance to subordinates and monitoring tools.	6	4	-	-
PC8. Learn reporting procedure & tools, report preparation to superiors and keeping the other teams informed.	6	4	-	-
PC9. Analyse Documentations methods, record maintenance, report submission & compliances.	6	4	-	-
NOS Total	60	40	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0218
NOS Name	Plan, Organise & Monitor Scaffolding Safety Protocols
Sector	Cross Sectoral
Sub-Sector	
Occupation	SCAFFOLDING ENGINEERING & MANAGEMENT
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The assessment criteria given in the Qualification Pack Scaffold Design Engineer by the SSDF.
2. Assessments can be carried out with pen-paper as well as online.
3. Assessment will be carried by certified assessor and affiliated Assessment Agency only.
4. Each NOS for its Performance Criteria (PC) has been assigned marks proportional to its importance. Proportion of marks for Theory and Practical has been marked NOS wise.
5. Questions on practical & theory will be formed in such a way as to provide outcome on maximum Performance Criteria and in proportional way within the NOS.
6. The assessment for the theory part will be based on written questions (short question, multiple choice & viva, or a combination of them) created/approved by the SSDF.
7. The assessment for the practical part will be based on practical conducted for trainees. In case of remote/on-line assessments, the practical's can be carried through proctors or practical questions formulated based on pictorially represented logical questions (based on pictures of practical & logical steps) created/approved by the SSDF.



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8. The passing and grading criteria of each NOS & cumulative for QP will be as follows:-

- 70% or more than 70% - Grade A
 - 60% or more than 60% but less than 70% - Grade B
 - 50% or more than 50% but less than 60% - Grade C
 - Less than 50% - Grade Fail
- e. If an individual gets less than 50% and 35% or more in one NOS and; overall, 50% or more; individual will be considered pass with grade "C" irrespective of overall marks.
- f. Individuals getting less than 50% in more than one NOS and getting overall marks 50% or more in QP will be put in grade Fail.
9. Assessment will be for complete QP and the trainees can be given additional chance to appear in re-assessment for improvements, based on decision by SSDF.
10. In case a trainee fails in one or two NOSs, re-assessments in the failing NOSs can be allowed in the next 3 months from the date of publication of results. Otherwise, the failed candidate will have to re-appear in all NOSs.
11. Minimum overall duration of assessment will be six hours.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 50

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSD/VSQ/N0213.Scaffoldings & Specifications.	60	40	0	0	100	17



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSD/VSQ/N0214.Understanding Scaffold Drawings & Designs, Indian & International Standard Codes	60	40	0	0	100	13
SSD/VSQ/N0215.Scaffold Design & Drawings using the scaffold & Computer-Aided Design (CAD) system	60	40	20	0	120	17
SSD/VSQ/N0216.Calculation of loads in scaffold designs as per Indian & International Standard	60	40	10	0	110	13
SSD/VSQ/N0217.Analysis of Scaffold design using STAAD Pro as per applicable IS and International Codes	60	40	20	0	120	22
SSD/VSQ/N0218.Plan, Organise & Monitor Scaffolding Safety Protocols	60	40	0	0	100	9
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	380	270	50	-	700	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.